



Original article

A predictive model for triple negative optic neuritis patients: Identifying the subset of patients with favorable outcomes

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ABSTRACT

Background and objectives: A subset of optic neuritis (ON) patients remains seronegative for AQP4-IgG, MOG-IgG, and CSF oligoclonal bands (OCBs), complicating diagnosis and management. This study aimed to characterize these triple-negative ON (TNON) cases and compare them to MS, MOGAD, and NMOSD.

Methods: We retrospectively analyzed 103 patients with a first ON episode tested for AQP4-IgG, MOG-IgG, and OCBs, classifying them into MS (n=26), MOGAD (n=25), NMOSD (n=13), and TNON (n=39). TNON was further divided into evolving MS (evMS) and idiopathic TNON. A support vector machine (SVM) was used to classify TNON based on onset data.

Results: TNON patients showed less optic disc edema and radiological involvement than MOGAD and responded more poorly to steroids. However, TNON had lower relapse rates and less need for chronic treatment than MS or NMOSD. SVM analysis revealed that most evMS cases clustered with MS, while idiopathic TNON aligned more closely with MOGAD.

Conclusion: TNON is a heterogeneous entity encompassing both early MS and a distinct idiopathic subgroup. While clinical features alone could not differentiate these subgroups, our predictive model successfully stratified them at onset. Idiopathic TNON may reflect an undetected antibody-mediated process with a generally favorable prognosis.

1. Introduction

Optic neuritis (ON), a pivotal diagnostic challenge in neurology and ophthalmology, manifests typically as an acute, often painful, loss of vision predominantly affecting young adults (Bennett, 2019; Biousse and Newman, 2016). Additionally, patients might experience color vision deficits and visual field defects (Levin et al., 2019). Historically associated with multiple sclerosis (MS), the etiological landscape of ON has expanded with the recognition of neuromyelitis optica spectrum disorder (NMOSD) and myelin oligodendrocyte glycoprotein (MOG) antibody-associated disease (MOGAD) (Petzold et al., 2022; Gospe et al., 2021). ON can present as a clinically isolated syndrome (CIS), characterized by isolated inflammation of the optic nerve without clinical or radiological evidence of other central nervous system (CNS) involvement (Jendretzky et al., 2024). In such cases, the prognosis varies widely

and depends on several risk factors. Patients who test negative for oligoclonal bands (OCBs) in the cerebrospinal fluid (CSF) exhibit a lower likelihood of progressing to MS compared to those with positive OCB findings (Schwenkenbecher et al., 2017). Among patients with CIS, 67.5% have positive OCB (Dobson et al., 2013). ON is the most common initial symptom in MS, occurring in 22–25% of cases, and 68% of patients presenting with ON eventually develop MS over an average of 15.8 years (Pestalozza et al., 2005).

NMOSD and MOGAD are antibody-mediated disorders distinguished by the presence of pathogenic autoantibodies (Sato et al., 2014). Aquaporin-4 antibodies (AQP4) antibodies are a hallmark of NMOSD, central to its diagnosis and pathogenesis, driving astrocytic damage and inflammatory cascades (Wu et al., 2019). Similarly, MOG antibodies serve as a diagnostic biomarker for MOGAD, targeting MOG and contributing to demyelination and axonal injury (Corbali and Chitnis,

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2023). The detection of these antibodies is critical for diagnosis, differentiation from MS, and guiding therapeutic strategies (Rosenthal et al., 2020).

A substantial subset of ON cases test negative for AQP4 antibodies, MOG antibodies, and OCB, rendering these cases 'seronegative' (Sato et al., 2014). The absence of definitive biomarkers in these patients complicates the clinical approach, often leading to a generic classification of 'idiopathic' once other systemic and neurological causes are ruled out (Galletta et al., 2015). This ambiguity presents significant challenges for management and understanding of the patient's prognosis (Siuko et al., 2019).

In this study we included all patients that presented with ON as the initial presentation to the neurology department between 2008 and 2024. This group included patients who were diagnosed with MOGAD, NMOSD and MS and a subset of patients who we classified as triple negative ON (TNON). We described their clinical parameters and compared the TNON group to NMOSD, MOGAD and MS patients. Additionally, we examined the long-term prognosis of the TNON group, assessing their relapse rate and need for immunosuppression compared to the other three groups.

2. Methods

2.1. Patients

This retrospective study utilized existing medical records from patients admitted to the Neurology Department at Hadassah Medical

Center between 2008 and 2024. The study included patients who underwent testing for OCB, AQP4 antibodies, and MOG antibodies, with available test results. All patients experienced their first clinical attack as ON. Serum samples were analyzed for MOG and AQP4 antibodies using the Euroimmun commercial biochip immunofluorescence cell-based assay (IIFT: Myelin-oligodendrocyte glycoprotein, product no. 1156). Only patients diagnosed with inflammatory ON were included, following the exclusion of other potential causes of optic neuropathy, including toxic, metabolic, and ischemic etiologies. For patients who experienced their clinical attack prior to the availability of the cell-based assay (CBA) technique in 2016, serologic testing was performed retrospectively as soon as the assay became accessible; thus, their serologic status is retrospective in nature. Out of 1,120 patients who underwent anti-MOG, anti-AQP4, and OCB testing, 639 had ON. Of them, 103 presented with ON as their first clinical manifestation and had available clinical data. This included 13 patients with AQP4-positive NMOSD which met the 2015 diagnostic criteria (Wingerchuk et al., 2015), 25 patients with MOGAD diagnosed using the international diagnostic criteria (Banwell et al., 2023; Rechtman et al., 2024), and 26 patients diagnosed with MS at clinical onset according to the 2017 McDonald criteria (Thompson et al., 2018). Additionally, we identified 39 TNON patients who tested negative for AQP4 and MOG antibodies, as well as OCBs, and did not meet the criteria for MS during their first clinical attack (Fig. 1). The TNON group was further subdivided into patients who were eventually diagnosed with MS during the follow-up years (evMS) of the study and those who never met the criteria for MS, MOGAD, or NMOSD, remaining without a specific diagnosis (idiopathic

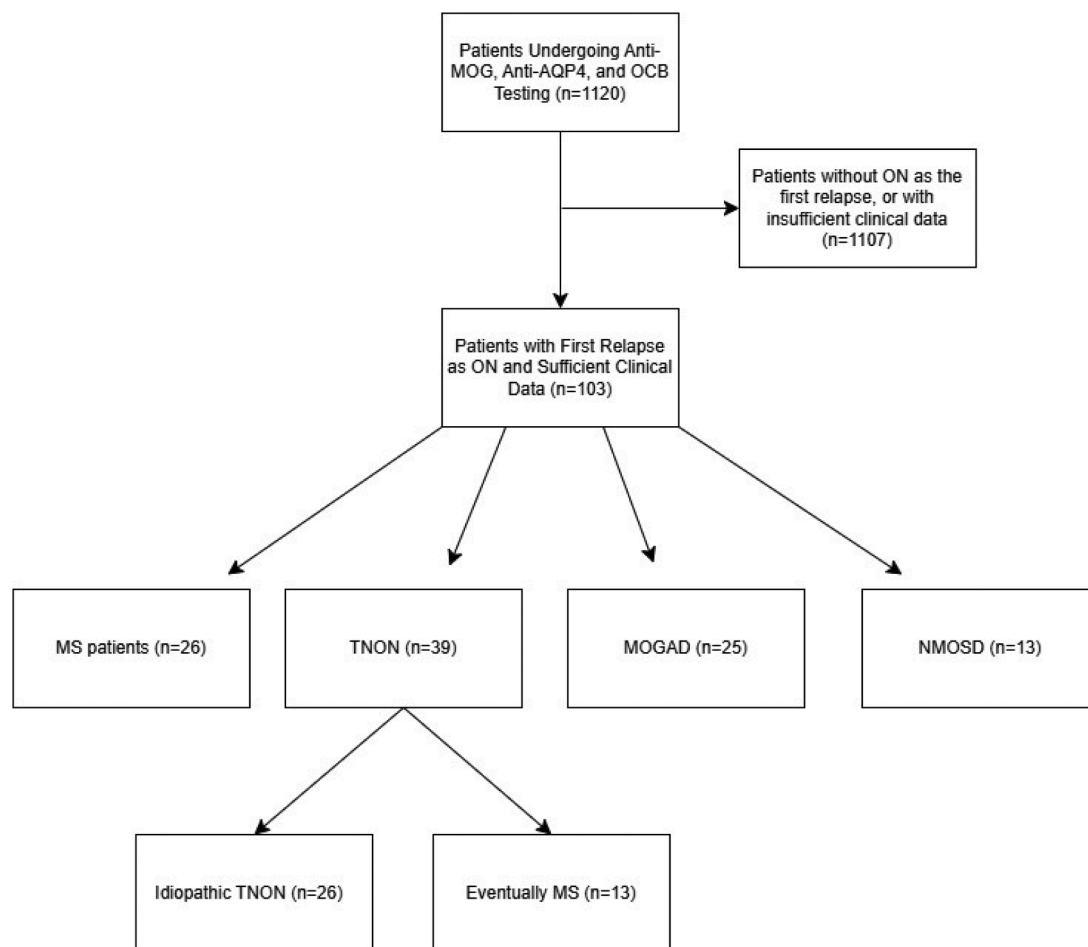


Fig. 1. Flowchart of the study

MOGAD: myelin oligodendrocyte glycoprotein antibody-associated disease; NMOSD: neuromyelitis optica spectrum disorders; MS: multiple sclerosis; TNON; Triple-negative Optic Neuritis; ON: Optic neuritis; MOG: myelin oligodendrocyte glycoprotein; AQP4: Aquaporin-4; OCB: Oligoclonal band.

TNON).

2.2. Data collection

At the time of presentation, we gathered a comprehensive dataset for each patient. This included demographic details such as sex and age at onset, as well as clinical and laboratory findings. Specifically, we recorded CSF lymphocyte count, glucose level, and protein concentration. We also assessed visual acuity, visual evoked potentials (VEP), relative afferent pupillary defect (RAPD), red desaturation, optic disc edema, and the presence of altitudinal visual field defects. Additionally, we documented optic nerve abnormalities on MRI, including hyperintense and enhancing lesions, as well as the presence of additional lesions on brain MRI. The status of other autoimmune disorders at presentation was also recorded.

We evaluated patient responses to first-line treatment with intravenous methylprednisolone (IVMP). Poor response was defined as limited response to a 3-day protocol of IVMP treatment or the need for second-line therapy or continuation of IVMP beyond this period. Disease progression was monitored through the number of relapses, the necessity for chronic treatment, and the appearance of new lesions on follow-up MRI.

2.3. Ethics

The study was approved by Hadassah Medical Organization's Ethics Committee (reference no. HMO-20-0644). Given the study design, the Hadassah Medical Organization's Ethics Committee determined that written consent was not required.

2.4. Visual fields

Visual fields were estimated using the automatic Humphrey perimetry test (SITA standard 24-2 protocol with stimulus 3; Carl Zeiss Meditec, Inc., Dublin, CA). Altitudinal visual field defect were defined as vision loss limited to either the upper or lower half of the visual field while respecting the horizontal meridian.

2.5. Visual evoked potentials latency

Standard full-field pattern reversal VEPs were recorded on a Bravo VEP device (Nicolet Biomedical) by a trained technician.

2.6. OCT

Peripapillary retinal nerve fiber layer (pRNFL) thickness was recorded by trained technicians using spectral-domain OCT (Spectralis, Heidelberg Engineering) with automatic real time (ART) function for image averaging.

2.7. MRI

Brain magnetic resonance imaging (MRI) scans included fluid-attenuated inversion recovery (FLAIR) and T1-weighted sequences with gadolinium contrast. Scans without lesions were classified as 'Normal'. Scans showing new or enlarging lesions on FLAIR-weighted images compared to a prior scan, and/or exhibiting gadolinium enhancement on T1-weighted images, were classified as 'active-MRI,' whereas all other scans were categorized as 'stable-MRI.'

Optic nerve involvement on MRI was classified as 'normal' or 'abnormal.' Abnormal optic nerves were defined as those exhibiting hyperintense lesions or gadolinium enhancement. Additionally, abnormal optic nerve involvement was further categorized as 'anterior' or 'posterior.' Anterior involvement included the intraocular and intraorbital segments, while posterior involvement encompassed the intracanalicular and intracranial segments.

2.8. Statistical analyses

For numerical data, a one-way ANOVA was utilized to determine if there were statistically significant differences in the means across the groups. Following the ANOVA, pairwise comparisons between each group were conducted using t-tests. For categorical data, chi-square tests were employed to assess the significance of differences.

A Support Vector Machine (SVM) model with a linear kernel was trained using data from the initial onset of MOGAD, NMOSD and MS groups. The data used for the training was brain MRI (Normal vs Abnormal), MRI of optic nerve (Normal vs Abnormal), sex, age of onset, ON side (unilateral vs bilateral), CSF test results (CSF lymphocytes, protein and glucose levels), RAPD (Yes vs No) and Optic disc edema (Yes vs No), response to steroids (Good vs Poor), and another autoimmune disorder (Yes vs No).

The TNON group served as the test set, divided into evMS and idiopathic TNON patients. A subset of TNON patients ($n=3$) was excluded due to missing values in at least one of the parameters required for the model. The model's performance was evaluated by classifying the test data and calculating the frequency of predictions. Additionally, the probability outputs were analyzed to assess the confidence of predictions by using the average probabilities for each class within the TNON group. To statistically compare the differences in probability distributions between MOGAD, NMOSD and MS as predicted by the model, Wilcoxon rank-sum tests were conducted.

3. Results

3.1. Comparative analysis of clinical, imaging, and CSF findings in TNON, MS, MOGAD, and NMOSD patients

We compared the clinical, imaging, and CSF tests between the TNON and the MS, MOGAD and NMOSD patients (Table 1).

The mean age of onset was similar between the TNON (31.12 ± 12.72) and MOGAD (29.10 ± 14.33) and MS (29.57 ± 10.59) groups, being slightly younger compared to the NMOSD (46.21 ± 27.06) group. There were no significant sex differences among the 4 groups (TNON: 29 Female (74%), 10 Male (26%); MOGAD: 15 Female (71%), 6 Male (29%); NMOSD: 11 Female (85%), 2 Male (15%), MS: 17 Female (65%), 9 Male (35%)). In terms of CSF tests, the TNON group exhibited lower CSF protein levels (38.89 ± 41.75 vs 116.09 ± 170.08 , $p=0.05$) and higher CSF glucose levels compared to the MOGAD group (66.92 ± 17.65 vs 58.79 ± 10.60 , $p=0.04$). Lymphocytes count in the CSF were significantly lower in the TNON compared to MS patients (1.92 ± 2.89 vs 8.00 ± 12.34 , $p=0.03$).

Regarding visual parameters at presentation, no significant differences were observed among the four groups in terms of visual acuity (TNON: 0.39 ± 0.43 ; MOGAD: 0.48 ± 0.27 ; NMOSD: 0.39 ± 0.28 ; MS: 0.43 ± 0.30), VEP latency in the affected eye (TNON: 133.50 ± 13.25 ; MOGAD: 134.80 ± 22.34 ; NMOSD: 127.75 ± 14.24 ; MS: 140.08 ± 19.90), or the fellow eye (TNON: 120.21 ± 14.72 ; MOGAD: 125.89 ± 16.82 ; NMOSD: 117.33 ± 2.31 ; MS: 120.89 ± 10.12). Additionally, there were no significant differences in the presence of an altitudinal visual field defect (TNON: 28 No (85%), 5 Yes (15%); MOGAD: 16 No (84%), 3 Yes (16%); NMOSD: 6 No (60%), 4 Yes (40%); MS: 19 No (86%), 3 Yes (14%)), the presence of RAPD (TNON: 7 No (18%), 32 Yes (82%); MOGAD: 4 No (19%), 17 Yes (81%); NMOSD: 2 No (14%), 12 Yes (86%); MS: 5 No (19%), 21 Yes (81%)), or red desaturation (TNON: 8 No (27%), 22 Yes (73%); MOGAD: 1 No (7%), 13 Yes (93%); NMOSD: 1 No (14%), 6 Yes (86%); MS: 5 No (21%), 19 Yes (79%)). RNFL of the affected eye was slightly lower in TNON compared to MOGAD (112.6 ± 39.6 vs 138.67 ± 49.52 , $p=0.06$), while no difference was found in the RNFL of the non-affected eye between the 4 groups (TNON: 97.97 ± 16.15 ; MOGAD: 98.83 ± 15.61 ; NMOSD: 88.86 ± 19.06 ; MS: 92.33 ± 14.05).

However, the TNON group exhibited significantly less optic disc

Table 1
Demographic, clinical and imaging parameters of TNON patients.

Variable	TripleNegative (n=39)	MOGAD (n=25)	NMOSD (n=13)	MS (n=26)	P_Value_MOGAD	P_Value_NMOSD	P_Value_MS
Age at Onset (Years)	31.12 ± 12.73	29.14 ± 12.54	46.22 ± 27.07	29.58 ± 10.60	0.57	0.07	0.60
Sex	F: 29, M: 10	F: 15, M: 6	F: 11, M: 2	F: 17, M: 9	1.00	0.70	0.62
Other Autoimmune Disorder	No: 33, Yes: 6	No: 21	No: 12, Yes: 1	No: 26	0.15	0.81	0.10
Visual Acuity	0.391 ± 0.427	0.479 ± 0.267	0.393 ± 0.281	0.426 ± 0.304	0.40	0.99	0.73
RAPD	No: 7, Yes: 32	No: 4, Yes: 17	No: 2, Yes: 11	No: 5, Yes: 21	1.00	1.00	1.00
Red Saturation	No: 8, Yes: 22	No: 1, Yes: 13	No: 1, Yes: 6	No: 5, Yes: 19	0.27	0.84	0.86
Optic Disc Edema	No: 27, Yes: 12	No: 6, Yes: 15	No: 8, Yes: 5	No: 19, Yes: 7	0.01	0.86	0.96
Altitudinal Field View	No: 28, Yes: 5	No: 16, Yes: 3	No: 6, Yes: 4	No: 19, Yes: 3	1.00	0.21	1.00
VEP Affected Eye	133.50 ± 13.25	134.80 ± 22.34	127.75 ± 14.25	140.08 ± 19.90	0.87	0.50	0.33
VEP Fellow Eye	120.21 ± 14.72	125.89 ± 16.82	117.33 ± 2.31	120.89 ± 10.12	0.42	0.50	0.90
RNFL Affected Eye	112.6±39.6	138.67±49.52	117.62±44.27	100.58±22.70	0.06	0.77	0.14
RNFL Non-Affected Eye	97.97±16.15	98.83±15.61	88.86±19.06	92.33±14.05	0.872	0.271	0.16
Lymphocytes in CSF	1.92 ± 2.89	6.38 ± 18.00	25.92 ± 64.12	8.00 ± 12.40	0.27	0.22	0.03
Protein in CSF	38.89 ± 41.75	116.10 ± 170.09	51.39 ± 38.00	33.48 ± 11.30	0.05	0.33	0.46
Glucose in CSF	66.92 ± 17.65	58.79 ± 10.603	69.04 ± 18.54	73.89 ± 25.49	0.04	0.72	0.51
OCB	Negative: 37, Positive: 0	Negative: 21, Positive: 0	Negative: 13, Positive: 0	Negative: 3, Positive: 22	1	1	<0.0001
Optic Nerve MRI Abnormality	Abnormal: 16, Normal: 23	Abnormal: 16, Normal: 5	Abnormal: 3, Normal: 10	Abnormal: 11, Normal: 15	0.02	0.41	1.00
Optic Nerve Lesion Distribution	Anterior: 8, Posterior: 8	Anterior: 12, Posterior: 2	Anterior: 1, Posterior: 2	Anterior: 7, Posterior: 3	0.09	1.00	0.55
MRI Head Abnormality	Abnormal: 19, Normal: 19	Abnormal: 7, Normal: 14	Abnormal: 11, Normal: 2	Abnormal: 23, Normal: 3	0.34	0.06	0.00
Spinal MRI Abnormality	Abnormal: 4, Normal: 19	Abnormal: 0, Normal: 12	Abnormal: 1, Normal: 10	Abnormal: 8, Normal: 12	0.41	1.00	0.08
Steroid Response	Good: 27, Poor: 12	Good: 20, Poor: 1	Good: 8, Poor: 5	Good: 19, Poor: 7	0.05	0.86	0.96
New Lesions at Follow Up	No: 22, Yes: 8	No: 14, Yes: 2	No: 8, Yes: 4	No: 12, Yes: 9	0.46	0.96	0.37
Relapse Number	0.50 ± 0.48	1.10 ± 1.25	0.31 ± 0.30	0.86 ± 0.88	0.04	0.12	0.06
Chronic Treatment	No: 27, Yes: 12	No: 13, Yes: 7	No: 4, Yes: 9	No: 7, Yes: 16	0.97	0.03	0.01

edema compared to the MOGAD group (TNON: 27 No (69%), 12 Yes (31%); MOGAD: 6 No (29%), 15 Yes (71%); $p = 0.006$).

The imaging findings revealed that the TNON group had less radiological evidence of optic nerve involvement compared to the MOGAD group (TNON: 23 normal (59%), 16 abnormal (41%); MOGAD: 5 normal (24%), 16 abnormal (76%); $p = 0.02$). Regarding the spatial distribution of optic nerve involvement in the TNON group, no clear localization of optic nerve involvement was observed (8 anterior (50%), 8 posterior (50%)).

The proportion of abnormal brain MRIs in the TNON group was comparable to that in the MOGAD group (TNON: 19 normal (50%), 19 abnormal (50%); MOGAD: 14 normal (67%), 7 abnormal (33%); $p = 0.34$) but slightly lower than in the NMOSD group (NMOSD: 2 normal (15%), 11 abnormal (85%); $p = 0.06$). Furthermore, the TNON group had a significantly lower prevalence of abnormal MRIs compared to the MS group (MS: 3 normal (12%), 23 abnormal (88%); $p = 0.004$). Furthermore, TNON group have slightly less abnormal spinal MRI compared to MS (TNON: 4 abnormal (17%), 19 Normal (83); MS: 8 abnormal (40%), 12 normal (60%); $p=0.08$).

Regarding the response to IVMP in the acute setting, the TNON group exhibited a poorer response compared to the MOGAD group (TNON: 27 good (69%), 12 poor (31%); MOGAD: 20 good (95%), 1 poor (5%); $p = 0.05$). During follow-up, TNON patients had a significantly lower relapse rate compared to the MOGAD group (0.50 ± 0.48 vs. 1.10 ± 1.25 ; $p = 0.04$) and a slightly lower relapse rate compared to the MS group (0.50 ± 0.48 vs. 0.86 ± 0.88 ; $p = 0.06$). Additionally, TNON patients received less chronic treatment compared to those with NMOSD (TNON: 27 no (69%), 12 yes (31%); NMOSD: 4 no (31%), 9 yes (69%); $p = 0.03$) and MS (TNON: 7 no (30%), 16 yes (70%); MS: 3 no (12%), 23 yes (88%); $p = 0.007$). Regarding new lesion development, no significant differences were observed among the four groups (TNON: 22 no (73%), 8 yes (27%); MOGAD: 14 no (88%), 2 yes (13%); NMOSD: 8 no (67%), 4 yes (33%); MS: 12 no (57%), 9 yes (43%)). Similarly, there was no significant difference in the prevalence of concomitant autoimmune

disorders (TNON: 33 no (85%), 6 yes (15%); MOGAD: 21 no (100%), 0 yes (0%); NMOSD: 12 no (92%), 1 yes (8%); MS: 26 no (100%), 0 yes (0%)).

3.2. Classification and probability distribution analysis of TNON patients using machine learning models

We divided the TNON group into two distinct subgroups as described in the Methods section: 'idiopathic TNON' and 'evMS' (Table 2). The evMS group had a significantly higher proportion of MRIs classified as 'Abnormal' compared to the idiopathic TNON group (idiopathic TNON: 17 normal (68%), 8 abnormal (32%); evMS: 2 normal (15%), 11 abnormal (85%); $p = 0.006$). Additionally, evMS patients required more chronic treatment (idiopathic TNON: 24 no (92%), 2 yes (8%); evMS: 3 no (23%), 10 yes (77%); $p < 0.0001$) and had a significantly higher proportion of 'active MRI' findings at follow-up (idiopathic TNON: 19 not active (100%), 0 active (0%); evMS: 3 not active (27%), 8 active (73%); $p < 0.0001$). Furthermore, evMS patients exhibited a slightly higher proportion of abnormal optic nerve involvement on imaging compared to the idiopathic TNON group (idiopathic TNON: 18 normal (69%), 8 abnormal (31%); evMS: 5 normal (38%), 8 abnormal (62%); $p = 0.13$). Finally, evMS patients have significantly higher proportion of abnormal spinal MRI findings compared to the Idiopathic TNON group (evMS: 4 abnormal (19%), 17 normal (81%); Idiopathic TNON: 0 abnormal (0%), 12 normal (100%); $p=0.006$).

We trained a SVM model using clinical and imaging data collected only at disease onset from patients diagnosed with MOGAD, NMOSD, and MS. The TNON group was then used as a test set, with separate classification for evMS and idiopathic TNON patients.

Using this model, evMS patients were predominantly classified as MS, with 10 out of 12 cases assigned to the MS category, compared to only 2 cases classified as MOGAD. In contrast, idiopathic TNON patients were primarily classified as MOGAD, with 17 out of 24 cases assigned to MOGAD and only 7 cases classified as MS (Fig. 2a).

Table 2
Demographic, clinical and imaging parameters differences between idiopathic and evMS patients.

Variable	Idiopathic TNON (n=26)	evMS (n=13)	P_Value
Age at Onset (Years)	32.05 ± 12.36	29.11 ± 13.82	0.54
Sex	F: 20, M: 6	F: 9, M: 4	0.90
Other Autoimmune Disorder	No: 21, Yes: 5	No: 12, Yes: 1	0.64
Visual Acuity	0.41 ± 0.47	0.35 ± 0.29	0.67
RAPD	No: 6, Yes: 20	No: 1, Yes: 12	0.46
Red Saturation	No: 5, Yes: 15	No: 3, Yes: 7	1.00
Optic Disc Edema	No: 16, Yes: 10	No: 11, Yes: 2	0.27
Altitudinal Field View	No: 20, Yes: 4	No: 8, Yes: 1	1.00
VEP Affected Eye	133.50 ± 14.29	133.50 ± 12.61	1.00
VEP Fellow Eye	122.50 ± 16.27	114.50 ± 9.15	0.27
RNFL Affected Eye	118.08±46.07	101.58±18.53	0.14
RNFL Non-Affected Eye	100.17±18.16	93.75±10.81	0.20
Lymphocytes in CSF	1.21 ± 2.00	3.23 ± 3.81	0.09
Protein in CSF	39.32 ± 49.77	38.09 ± 21.89	0.92
Glucose in CSF	66.45 ± 12.31	67.85 ± 25.94	0.86
Optic Nerve MRI Abnormality	Abnormal: 8, Normal: 18	Abnormal: 8, Normal: 5	0.13
Optic Nerve Lesion Distribution	Anterior: 4, Posterior: 4	Anterior: 4, Posterior: 4	1.00
MRI Head Abnormality	Abnormal: 8, Normal: 17	Abnormal: 11, Normal: 2	0.01
MRI Spinal Abnormality	Abnormal: 0, Normal: 12	Abnormal: 4, Normal: 17	0.006
Steroid Response	Good: 17, Poor: 9	Good: 10, Poor: 3	0.71
New Lesions at Follow Up	No: 19	No: 3, Yes: 8	<0.0001
Relapse Number	0.41 ± 0.38	0.68 ± 0.61	0.18
Chronic Treatment	No: 24, Yes: 2	No: 3, Yes: 10	<0.0001

The reliability of this classification was further supported by the average SVM probabilities of evMS, which strongly favored MS (ANOVA; $p < 0.0001$) compared to both MOGAD (0.60 ± 0.19 vs. 0.21 ± 0.25 , $p < 0.0001$) and NMOSD (0.60 ± 0.19 vs. 0.19 ± 0.11 , $p < 0.0001$) patients. Furthermore, the average SVM probabilities of idiopathic TNON showed a significant preference for MOGAD (ANOVA; $p < 0.0001$) compared to both MS (0.46 ± 0.23 vs. 0.33 ± 0.13 , $p = 0.03$) and NMOSD (0.46 ± 0.23 vs. 0.20 ± 0.14 , $p < 0.0001$) (Fig. 2b).

Moreover, evMS patients had significantly higher probabilities of being classified as MS compared to idiopathic TNON (0.33 ± 0.13 vs. 0.60 ± 0.19 , $p < 0.0001$), while idiopathic TNON patients had higher probabilities of being classified as MOGAD (0.46 ± 0.23 vs. 0.21 ± 0.25 , $p = 0.005$). There was no significant difference in the probabilities of being classified as NMOSD between evMS and idiopathic TNON (0.19 ± 0.11 vs. 0.20 ± 0.14 , $p = 0.79$) (Fig. 2c).

4. Discussion

This study provides a comprehensive characterization of patients presenting with ON as their initial neurological manifestation, emphasizing differences in clinical, imaging, and CSF findings across patients diagnosed with MOGAD, NMOSD, MS, and those classified as TNON. The TNON group exhibited distinct clinical and radiological features, raising important considerations regarding its pathophysiology and long-term prognosis.

Given that both elevated CSF protein and lymphocyte levels are markers of increased inflammation, the significantly lower CSF protein levels in TNON compared to MOGAD, and the lower CSF lymphocyte count compared to MS, suggest a less pronounced inflammatory response in TNON. Despite optic disc edema being less frequent in the TNON group compared to MOGAD, visual acuity, VEP latency, and visual field defects did not differ significantly among the four groups at presentation, indicating comparable degrees of initial optic nerve dysfunction.

The response to first-line IVMP was notably poorer in TNON

compared to MOGAD. This finding emphasizes the potential differences in immune pathophysiology and steroid sensitivity between these conditions. Additionally, TNON patients had significantly lower relapse rates than MOGAD and MS, suggesting a more monophasic or less aggressive disease course. This was further supported by the lower proportion of TNON patients requiring chronic immunosuppressive therapy compared to NMOSD and MS.

A key contribution of this study is the subdivision of TNON into idiopathic TNON and evMS cases. The evMS group exhibited a significantly higher prevalence of abnormal brain MRI findings, an increased need for chronic immunosuppression, and more active MRI lesions at follow-up, distinguishing them from idiopathic TNON patients. These findings suggest that a subset of TNON cases represents an early stage of MS.

The recognition of two distinct TNON subgroups with differing prognoses emphasises the need for early diagnostic tools. Our SVM model accurately distinguished these groups using only clinical data from the initial ON episode. Most evMS cases were classified as MS, while idiopathic TNON was predominantly classified as MOGAD, with significantly higher SVM probabilities favoring MOGAD over MS and NMOSD. This suggests idiopathic TNON may represent a distinct autoimmune condition, or an antibody-mediated process not yet identified. One possibility is the presence of an undetected antibody targeting the optic nerve or CNS, similar to how MOG-IgG and AQP4-IgG discoveries redefined ON subtypes. Alternatively, these cases may involve low-titer MOG antibodies not captured by current assays. These findings highlight the need for novel biomarkers and refined testing to guide early diagnosis and treatment strategies in triple-negative ON.

We also examined spinal MRI abnormalities across groups. The TNON group demonstrated a lower frequency of spinal cord lesions compared to the MS group, though this difference did not reach statistical significance. Within the TNON subgroup, however, evMS patients exhibited a significantly higher rate of abnormal spinal MRI findings than idiopathic TNON patients. These results suggest that spinal involvement may serve as a useful indicator of early MS, even in the absence of overt spinal symptoms. Based on these findings, we recommend that spinal MRI be considered in patients presenting with optic neuritis when MS is suspected, even if there are no clinical signs of myelopathy. Notably, spinal MRI was not included in our predictive model, as a considerable proportion of patients did not undergo this examination.

Our model incorporates clinical and laboratory features beyond those considered in the proposed 2024 McDonald criteria, which focus on radiological markers such as the central vein sign and optic nerve involvement. Unlike the criteria, which require spinal MRI, our model relies solely on brain MRI, enhancing its utility in settings where spinal imaging is unavailable. Notably, only 1 of the 13 evMS patients in our cohort would have been diagnosed with MS at onset using the new criteria, highlighting our model's added value in enabling earlier identification.

Limitations of our study include the relatively small cohort, which reflects both the rarity of this disorder and the need for comprehensive data from each patient. Additionally, MOG antibodies were not tested in the CSF, so it is possible that one of the TNON patients may have MOGAD. However, it is unlikely that this accounts for more than a few cases, as CSF-positive MOGAD without serum positivity is a rare phenomenon. Despite these limitations, we believe that this study provides an important step toward more accurate classification of TNON patients.

5. Conclusion

In conclusion, our findings indicate that this newly identified TNON entity essentially comprises two distinct subgroups: patients who will eventually develop MS, and a "true" TNON subgroup, which we defined as idiopathic TNON. Although individual clinical parameters alone could not differentiate these two subgroups, our model effectively

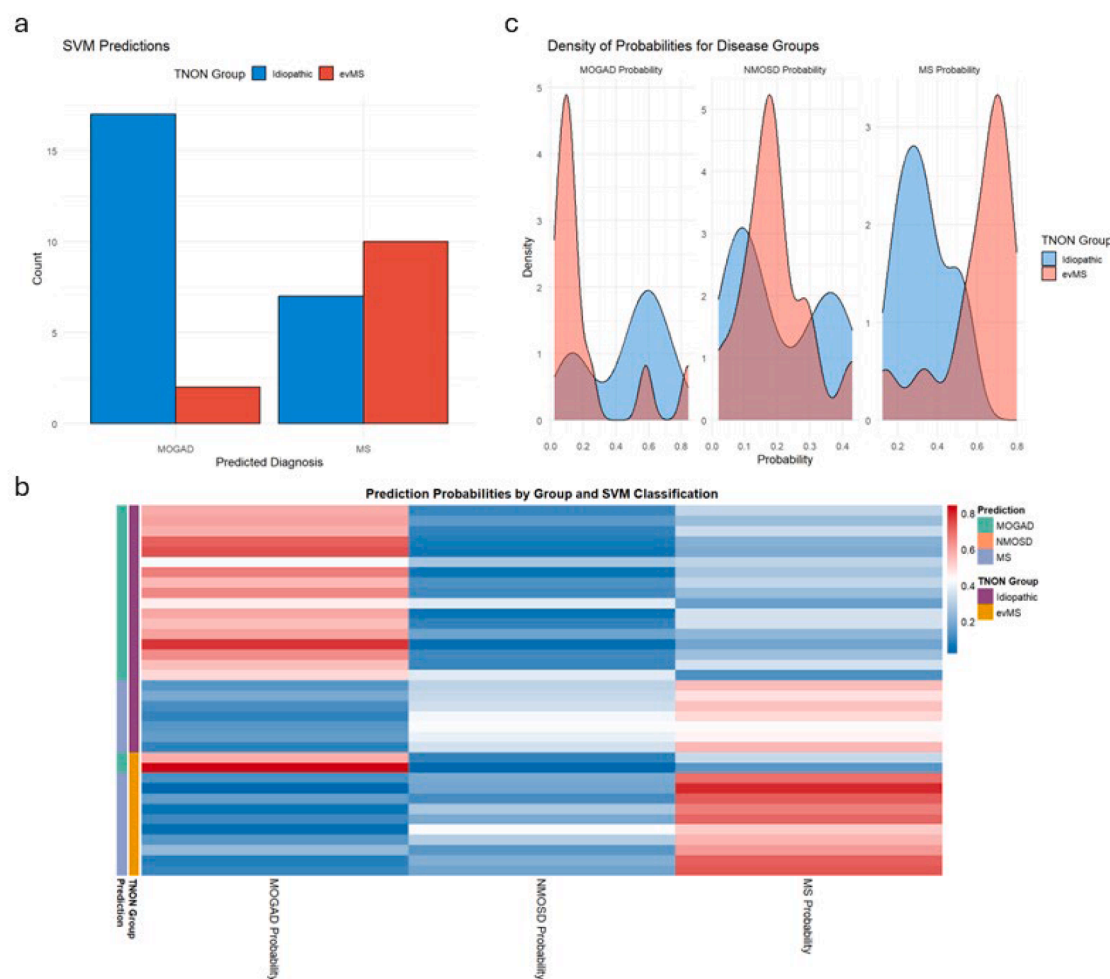


Fig. 2. Idiopathic TNON Have a greater probability of being classified as MOGAD

Fig. 2. (a) evMS (10 MS, 2 MOGAD) and idiopathic TNON (7 MS, 17 MOGAD) SVM model classification. (b) Heatmap of individual-level SVM-predicted probabilities for being classified as MOGAD, NMOSD, or MS. Rows represent patients and are annotated by TNON subgroup and predicted diagnosis. Columns represent disease probabilities (c) Density plots showing the distribution of predicted probabilities for each disease group (MOGAD, NMOSD, and MS) stratified by TNON subgroup. MOGAD: myelin oligodendrocyte glycoprotein antibody-associated disease; NMOSD: neuromyelitis optica spectrum disorders; MS: multiple sclerosis; TNON; Triple-negative Optic Neuritis; evMS: eventually MS.

predicted patients who progressed to MS. We propose that idiopathic TNON represents a distinct subset of patients whose underlying pathology may be driven by an unidentified antibody. The notable classification bias toward MOGAD observed in our SVM model further supports an immune-mediated mechanism, even though this subgroup currently lacks an identified biomarker. Validation of our SVM model could help identify true TNON patients more accurately, thereby facilitating the discovery of the underlying autoantibody.

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CRediT authorship contribution statement

Ariel Rechtman: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Tal Friedman-Korn:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Sol Zamir:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Omri Zveik:** Writing – original draft, Visualization, Investigation, Data curation.

Netta Levin: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Adi Vaknin-Dembinsky:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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